Atomic elementary flux modes explain the steady state flow of metabolites in large-scale flux networks

aEFMs explain metabolite flows

These are two separate affiliations. The uOttawa should be 2. Even though we're both part of both, they need to be listed separately.

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Would have expected commas not forward slash?

[illegible]g author.

Flux network/Elementary flux mode/flow decomposition/Markov chain/atom mapping

Abstract

Careful! Do we do any work on "reaction rates"?

Steady state flux[illegible] cellular activity under metabolic homeost[illegible] changes in reaction rates remains challenging in large-scale metabolic netw[illegible] pproaches such as elementary flux mode (EFM) anal[illegible] mal[illegible] to [illegible] combinatorial explosi[illegible]ys and ambigu[illegible]n EFMs. [illegible]sent an alternati[illegible] etabolic flu[illegible] eady state fl[illegible] molecular constituen[illegible] atomic EFMs and describe a fast and scalable m[illegible] to both enumerate these pa[illegible]nd uniquely identify their weights. Using our approach, we enumerate carbon and nitrogen EFMs in five genome-scale metabolic models and comput[illegible] their correspo[illegible]ghts in a HepG2 liver cancer cell line. Our results systematically c[illegible]acterize source m[illegible]emodelling and reveal the most active pathways of HepG2 glutami[illegible]etabolism. We thus anticipate our method will be useful to quantify cellular metab[illegible]-throughpu[illegible] fluxomic analyses become increasingly common.

Maybe "in terms of EFMs" or "onto EFMs"? I worry "on EFMs" might not be understood.

Or atomic?

better: "as atomic elementary flux modes (AEFMs)" (or aEFMs if, if you really want that "a" to be lower case.)

"fast and scalable" is questionable. "faster and more scalable" is not so questionable. Depends if w[illegible] want to be accurate or bo[illegible] in the abstract. Although "faster and more sca[illegible]

aEFMs! Keep pounding that new terminology. Branding!

remove "thus"

"their weights" may be too technical for abstract. how about: "and compute the aEFM decomposition of metabolic fluxes measured (or estimated?) in the HepG2 liver cancer cell line.

Then again, I just had a totally different thought about how to make parsimony-based EFM analysis more feasible. Maybe we could discuss sometime.

Introduction

Recent technological and computational advancements are leading to a rise in metabolic flux data (**?**). These fluxes quantify the biochemical reaction rates of metabolites interconverting within a cell or moving across subcellular compartments. From a systems biology perspective, fluxes are the most downstream readout of the metabolic phenotype, resulting from interactions between the genome, transcriptome, proteome, and metabolome. Hence, 'fluxomics' (Emwas et al, 2022) is emerging as a field dedicated to quantifying the totality of fluxes within biological systems.

Current experimental and computational methods generally estimate fluxes under metabolic homeostasis. Under this assumption, the production and consumption of metabolites are balanced over time, and the fluxes are said to be under steady state. Techniques such stable isotope tracing analysis (SITA) experimentally determine steady state fluxes by culturing cells in labelled substrates and quantifying their incorporation within endogenous metabolites (Buescher et al, 2015; Antoniewicz, 2018). Using this method, fluxes have been measured in major metabolic subsystems such as glycolysis, pentose phosphate pathway, and citric acid cycle (Orth et al, 2010; Ahn et al, 2017; Wu et al, 2019). Many computational methods have also been proposed to infer steady state fluxes from transcript, protein, and metabolite abundance data (Uematsu et al, 2022; González-Arrué et al, 2023; Huang et al, 2023; Kaste & Shachar-Hill, 2023; Lee et al, 2023). Collectively, these approaches have been used to estimate and predict fluxes i[illegible]me-scale metabolic models (GEMs) involving hundreds to thousands of reactions ([illegible]rkaoui et al, 2022; Gelbach et al, 2024).

Why calling it '-omics' ?

Although much effort has been made to estimate st[illegible] less emphasis has been placed on developing methods to analyze this '-omics' data. Steady state fluxes are particularly unique compared to other '-omics' data because steady state implies conservation between metabo-

lite inflows and outflows over time. In a network with [illegible] and sink fluxes, for example, an increase in flux along one metabolic pathway sh[illegible]ther pathways to maintain mass conservation. For simple networks, one [illegible]e system would evolve by identifying the up and downstream reactions f[illegible]tic perturbation. For small-scale networks, it may also be possible to enumerate all possible ways that fluxes could change in response to a perturbed reaction. Yet in highly-connected metabolic networks at the genome-scale, there exists no hierarchical relationship between reactions nor the concept of upstream and downstream reactions. A change in one reaction rate, let along multiple differentially active fluxes, may impact a multitude of neighbouring reactions within the metabolic network. Hence, there is a great need for more sophisticated computational tools to analyze large-scale metabolic flux datasets.

These claims need citations.

This problem of understanding the flow of metabolites within steady state flux networks has historically been addressed through the concept of elementary flux modes (EFMs). EFMs are a pathway-based approach to analyze metabolic networks and are defined as minimal sets of biochemical reactions that maintain steady state flux (Schuster & Hilgetag, 1994). This minimal property specifies that an EFM cannot be decomposed into two or more smaller pathways carrying steady state flux. An attractive property of EFMs is that any set of steady state fluxes in a network can be explained as a positive, linear combination of EFMs (Schuster et al, 1999). While EFMs are a mechanistic explanation of how metabolites travel through networks, the number of EFMs scales combinatorially as a function of network size. It thus remains computationally infeasible to enumerate EFMs in genome-scale networks after over a decade of algorithmic advancements (e.g. Gagneur & Klamt (2004); Terzer & Stelling (2008); Buchner & Zanghellini (2021)). The number of EFMs also poses a challenge for explaining network fluxes in terms of their EFMs. When there are more EFMs than linearly independent fluxes, there exist multiple ways to assign EFM weights that reconstruct the observed network fluxes. Consequently, structural analyses of EFMs and down-

stream analyses of their weights have been limited to small-scale metabolic networks (Schwartz & Kanehisa, 2006).

grammar

In our previous work, we took the first steps to solve the EFM flux decomposition for the special case of unimolecular reaction networks under a Markov constraint. We modelled a hypothetical particle moving between metabolites under the assumption that the transition probability of each reaction was proportional to the observed steady state flux. By expanding the notion of Markov chain state to include sequences of metabolites, we proposed an algorithm to efficiently compute simple cycle probabilities corresponding to a given EFM. We called this construct the *cycle-history Markov chain* (CHMC) and proved that these EFM probabilities could be scaled to weights that fully reconstructed the flux along each reaction (Chitpin & Perkins, 2023).

In this work, we generalize our CHMC method to operate on any type of metabolic model including those involving multispecies reactions. We do so by proposing a new type of *atomic EFM* which traces the possible steady state flow of an individual atom through a metabolic network as it remodels through various internal metabolites. Similar to standard (or molecular) EFMs, the totality of atomic EFM weights for a given source metabolite will reconstruct its observed mass flow entering the network. We first introduce our framework for enumerating and uniquely identifying atomic EFM weights. Using a state-of-the-art atom mapping algorithm, we show how to generate atomic transition graphs from which we construct atomic CHMCs. Using our approach, we enumerate atomic EFMs in genome-scale metabolic networks considered infeasible by state-of-the-art molecular EFM enumeration programs. We further perform a structural analysis of atomic EFMs to characterize nutrient source remodelling. Lastly, we analyze a single set of atomic EFM weights estimated from a HepG2 liver cancer cell line to quantify glutamine metabolism. Our predicted pathways of glutamine metabolism not only corroborate experimental studies in literature (**?**), but predict glutamine remodelling through a non-canonical TCA pathway recently validated in mouse

embryonic stem cells (Arnold et al, 2022).

Results

Tracing atomic flows in genome-scale models

We first describe what standard 'molecular' EFMs look like in metabolic networks and provide intuition why enumerating and decomposing fluxes onto these pathways remains a fundamental problem in literature. Figure 1a shows an example molecular EFM which shows how the substrates glucose and ATP are consumed to generate product metabolites G3P, DHAP, and ADP. Multiple source and sink fluxes are required to balance the internal metabolite stoichiometry, leading to several branching and converging paths within the molecular EFM. In large [illegible] molecular EFMs become increasingly complex with numerous source and [illegible] stoichiometrically balance participating metabolites and metabolic byproducts ([illegible] hydrogen ions). This complexity leads to an intractably large number of pathways that may only differ by reaction subsets that are required to mass-balance the metabolic byproducts. Choosing which metabolic byproducts to remove may, in part, solve this problem. However, choosing what metabolites to prune from the network is often subjective and not guaranteed to sufficiently reduce the total number of molecular EFMs.

Remove "which". And remove "what" from next sentence.

Given many metabolic flux networks are generated to study nutrient catabolism (Zamboni et al, 2015), we argue for studying source metabolite-derived atomic EFMs to reduce the combinatorial number of molecular EFMs. We refer to these pathways as *atomic EFMs* which trace the minimal, steady state flow of an individual atom within a metabolic network. An example carbon EFM is highlighted in Figure 1b which shows how the C-6 glucose carbon is remodelled during the first steps of glycolysis. As atoms are never split apart or fused together in metabolic reactions, these atomic

EFMs always correspond to linear pathways, regardless of t[illegible]mber of participating substrates/products in each reaction. [illegible]o desirable because it enables us to enumerate and compu[illegible]sly described cycle-history Markov chain (CHMC) method for decomp[illegible] EFMs derived from a unimolecular reaction network (Chitpin & Perkins, 2023).

Eh... they can be loops too. Are loops "linear pathways"? And we don't want the reader to think we don't allow branches in pathways either. Clarify this!

Figure 1c outlines our pipeline to enumerate and identify atomic EFM weights. In step 1, we require as inputs the metabolic flux network defined by a set of metabolites and reactions encoded as a stoichiometry matrix, the SMILES strings for each metabolite, and (optionally) the steady state fluxes. From the stoichiometry matrix and SMILES strings, we construct reaction SMILES strings which are imputed into the atom mapping program RXNMapper (Schwaller et al, 2021) to trace the position of each atom within a reaction equation. This atom mapping data is used to construct an atomic transition graph in step 2, which traces the atomic position of an individual atom from a given source metabolite entering the network. We set the transition probabilities equal to the atomic flux through that reaction. For example, 10mM of input glucose flux would result in 10mM of atomic flux through each glucose carbon. From each atomic transition graph, we construct a corresponding atomic CHMC to enumerate all atomic EFMs involving that source metabolite atom in step 3. The atomic EFM weights are finally computed in step 4 by a steady state analysis of the CHMC. If no steady state fluxes are imputed, only the atomic EFMs are returned in step 3 without any associated weights.

Atomic EFMs are computationally tractable in large-scale networks

Given the major computational challenge of enumerating molecular EFMs in networks with > 100 metabolites and reactions, we first sought to test whether it was feasible to enumerate atomic EFMs in GEMs. We selected five genome-scale networks across several organisms in ascending

Where is that Table 1? And/or can we talk about the statistics in the text, to give the reader a sense of their size and complexity? Also, can these things be visualized? E.g. in cytoscape? It doesn't have to look pretty, but just again to convey their complexity?

order of network size. After pre-processing these dat

SMILES strings, we constructed atomic CHMCs rooted

all source metabolites. These atoms were chosen f their metabolic significance in comparison with oxygen or hydrogen atoms. The resulting summary statistics describing the GEMs and atomic EFMs are presented in Table 1. For comparison, we attempted to enumerate the molecular EFMs using FluxModeCalculator, which is the most advanced EFM enumeration program (van Klinken & Willems van Dijk, 2016).

Figure 2a shows the total number of atomic and molecular EFMs enumerated by either program on the five GEMs. Using our atomic CHMC pipeline, we enumerated between $10^3 - 10^6$ carbon and nitrogen EFMs within each GEM. In contrast, FluxModeCalculator failed to enumerate molecular EFMs in all but the two smallest E. coli core and iAB RBC 283 datasets, returning $10^2 - 10^4$ times more molecular-versus-atomic EFMs. We observed that many of these molecular EFMs only differed by how their metabolic byproducts were stoichiometrically balanced. By focusing on an atom of interest, our atomic CHMC approach greatly reduced the number of EFMs to biologically relevant pathways involving nutrient source remodelling. Furthermore, our approach was significantly faster than the state-of-the-art program FluxModeCalculator. Our serial benchmarking showed atomic CHMC computation times to be $10^3 - 10^6$ times faster than FluxModeCalculator. Although FluxModeCalculator does benefit from multiple threads, it was unable to enumerate molecular EFMs in the remaining three GEMs. We determined that FluxModeCalculator 16 threads was the optimal number for the E. coli core and iAB RBC 283 dataset (see Expanded Views 1) with a 7.4-8 times performance speedup. Nevertheless, our atomic CHMC approach, running in serial, still performed $10^2 - 10^4$ faster than FluxModeCalculator running (optimally) in parallel.

We next investigated whether there was a relationship between atomic CHMC enumeration times and the number of atomic EFMs. For molecular EFM enumeration algorithms, it is well

Log-linearly? No! For two reasons. Think about this again.

established [illegible] the number of metabolites or reactions within the model (Ullah et al, 2020). Figure 2c shows that atomic CHMC enumeration times scale log-linearly as a function of the number of atomic EFMs. While there is still a combinatorial explosion of pathways, the CHMCs are independent from one another and their calculations can be performed separately. However, our software to compute atomic CHMCs is also parallelized at the level of individual atomic CHMCs. Additional threads may be specified to speed up CHMC state space and EFM enumeration. For the top 33 largest carbon CHMCs in the HepG2 dataset, we re-enumerated their EFMs with varying numbers of threads. Figure 2d reports the algorithm scaling, where we observed a median speedup of 6.9 times when using 32 threads. We found that the greatest speedups were associated with the largest atomic CHMCs. The largest atomic CHMC, for example, corresponded to a glutamine carbon that transited 211,456 atomic EFM and exhibited an 8.0 times speedup when enumerated with 32 threads.

Structural analysis of atomic EFMs

pathways

Having systematically enumerated all carbon and nitrogen EFMs across the five GEMs, we [illegible] to characterizing their structural properties. Molecular EFMs can be divided into pathway that span source and sink metabolites or internal loops within the metabolic network such as pairs of reversible reactions. Atomic EFMs can be further classified into those that transit sets of distinct metabolites or those that revisit the same metabolite, albeit in a different atomic position, at least once. An example of a common source-to-sink atomic EFM with metabolite revisitations is a carbon EFM tracing glucose remodelling through two turns of the TCA cycle; on the second turn, the position of the carbon atom shifts within the TCA intermediates, resulting in different isotopomers and therefore a metabolite revisitation under our definition.

To obtain a broad understanding of the types of atomic EFMs characterizing the five GEMs,

Why does a,b legend have double bars for each type instead of single bars?
Plus I'm still not keen on these horizontal bar graphs.
Consider stacked bar graph?
Also, no panel letters visible?

we categorized their [illegible] these four classes. Figure 3a-b shows the proportions of each atomic EFM classes within each GEM. In gene[illegible] bon EFMs describe nutrient source metabolism and involve source-to-si[illegible] tabolite revisitations. As the number of metabolites and reactions increased, we observed a broad decrease in the number of looped carbon and nitrogen EFMs. Looped carbon EFMs with metabolite revisitations, in particular, were extremely rare across all datasets. Only 0.4% of nitrogen EFMs within the iIT341 and iSB619 GEMs involved internal loops with metabolite revisitations.

Is that the order the models are listed in? Order is different from Figure 2. Plus, numbers are not visible on this chart. Percentages are.

Figure 3c-d shows the distribution of atomic EFM lengths across the five GEMs. The carbon EFMs from the four BiGG database models tended to be quite short with few pathways exceeding 50 reactions. This finding suggested the metabolic models exhibited weak connectivity between substrates, even in the larger iSB619 dataset which contained the greatest number of metabolites and reactions among all five GEMs. Surprisingly, the HepG2 carbon EFM lengths exhibited a Gaussian distribution with mean of 62.8 and standard deviation of 15.9. Perturbing the average carbon EFM in the HepG2 model would likely impact fluxes through approximately 62 other reactions, which highlights the high degree of connectivity within this network. We also noted the distribution of nitrogen EFM lengths was much broader across all datasets with the exception of the E. coli core model which contained only thirty nitrogen EFMs.

We next quantified how distinct atomic elements within given source metabolites could be metabolized via different EFMs within the network. For each source metabolite, we computed the log ratio of source metabolite carbons or nitrogens with the greatest and fewest number of atomic EFMs across the five GEMs. Figure 3e-f shows these statistics plotted for carbon and nitrogen EFMs. We observed that 47.3% and 34.6% of source metabolites with at least one carbon and nitrogen, respectively, were metabolizable through different numbers of atomic EFMs. This ratio increased to 49.8% and 74.5% after removing metabolites with only a single carbon or nitrogen

atom (e.g. carbon dioxide, methanol, glycine). We found that many types of amino acids exhibited differential metabolic potential, where certain carbons could participate in over 10^4 times as many pathways versus others within the same source metabolite. Similar findings were observed for amino acids with multiple nitrogens, which exhibited a 10^2 difference in the number of nitrogen pathways.

A minority of carbon EFMs explains the majority of source metabolite remodelling

Glutamine is a non-essential amino acid that is highly metabolized in HepG2 and other cancer cell types (**?**). Given the single set of steady state fluxes in the HepG2 dataset, we computed glutamine-derived carbon EFM weights to study how glutamine was metabolized in this genome-scale model. From our HepG2 GEM, we found that input glutamine carbons could undergo a total of 393,160 EFMs. As these were too many pathways to individually analyze, we sought to identify the most dominant glutamine-derived carbon EFMs carrying the greatest carbon flow through the network.

Figure 4a shows the cumulative explained mass flow of each glutamine carbon as a function of their EFMs sorted by greatest to smallest explained mass flow. We capped the cumulative explained mass flow of each glutamine carbon at 99% because we observed that hundreds to hundreds of thousands of carbon EFMs collectively explained the remaining 1% glutamine mass flow. Our results indicate that the majority of glutamine carbon remodelling is explained by a surprisingly small number of glutamine EFMs. The top 5 ranked EFMs across each glutamine carbon collectively explained over half the total glutamine mass flow. Specifically, the top 5, 10, and 50 carbon EFMs collectively explained 51.5%, 63.3%, and 89.4% of the total glutamine carbon flow. These observations were consistent across carbon EFMs derived from other amino acids as well as glucose (see Expanded Views 2).

From the top 5 glutamine carbon EFMs, we constructed a metabolic subnetwork with metabolite/atom states collapsed onto distinct metabolites and weighted by the aggregated EFM weights. Figure 4b shows the resulting graph where the majority of glutamine carbon flows were distributed across just 26 metabolites and 49 reactions. Of these metabolites, only 4 corresponded to two disconnected pairs of metabolites cycling carbon flux between 5,10-methylene-THF/5,10-methenyl-THF, and GSH/S-sulfanylglutathione. We found the majority of glutamine remodelling occurred within the remaining 22 metabolites which partitioned into two distinct clusters connected by mitochondrial glutamate. One cluster corresponded to input glutamine carbons flowing into the network and ultimately leaving via cytosolic alanine. The second and much larger cluster, however, encoded numerous cyclic pathways remodelling glutamine through a number of TCA intermediates.

To better understand the glutamine carbon flows, we visualized the top 5 glutamine EFMs derived from each of its 5 carbons. Figure 4c-f shows the resulting Sankey diagams. Because the top five EFMs and weights were identical between glutamine backbone carbons, they are represented by a single diagram. Focusing on these backbone carbons in Figure 4c, we observed that the first, third, and fifth highest weighted EFMs corresponded to carbon remodelling through the malate-aspartate shuttle. The fourth largest EFM corresponded to forward TCA cycle activity. However, we noticed that the second largest EFM was moving carbons between citrate, oxaloacetate, and malate between the cytoplasm and mitochondria. Interestingly, this metabolic pathway corresponds to the recently discovered "non-canonical" TCA activity described by Arnold et al, who validated this pathway in mouse embryonic stem cells and C212 myoblasts by stable isotope-tracing analysis. The Sankey diagrams for the remaining glutamine carbons generally reflected simple cycles corresponding to different aspects of TCA activity. The top EFMs of glutamine carbon 3 in Figure 4d, for example, recapitulated non-canonical and canonical TCA activity, while the largest EFM in glutamine carbons 4 and 5 glutamate corresponded malate-aspartate shuttle activity. Nearly all EFMs

corresponded to cyclic pathways with the exception of a single source-to-sink EFM transforming input glutamine carbons into alanine in Figure 4e-f.

Discussion

Since the proposal of molecular EFMs, much effort has been made, with little success, to enumerate and decompose steady state metabolic fluxes onto these pathways. By defining EFMs with respect to individual atoms rather than molecules, we introduced the concept of atomic EFMs. As strictly linear pathways, atomic EFMs are easier to interpret than their molecular EFM counterparts, which may involve multiple inputs and outputs to balance molecular stoichiometries. Using the atom mapping program RXNMapper, we adapted our previously described CHMC method to efficiently enumerate carbon and nitrogen EFMs in five GEMs. Our structural analysis systematically characterized source metabolite remodelling at the atomic level. By comparing the number of EFMs across atoms within the same source metabolite, we revealed how certain amino acid display remarkable metabolic variability within their constituent atom types. Our EFM weight analysis of the HepG2 network revealed that the majority of glutamine carbons were remodelled via cyclic pathways regenerating TCA intermediates. Excitingly, our model also predicted non-canonical TCA activity which has not been previously described in HepG2 cells.

As a systematic method to explain the flow of metabolites, the problem of EFM enumeration has historically limited their applicability to small-scale networks. Many algorithmic advances enumeration have incrementally improved EFM enumeration, yet never feasibly beyond networks with > 100 metabolites and reactions (Kamp & Schuster, 2006; Terzer & Stelling, 2008; van Klinken & Willems van Dijk, 2016). EFM enumeration is generally regarded as unfeasible for large-scale networks—so much so that numerous methods have been developed to enumerate a subset of EFMs

constrained by pathway length (de Figueiredo et al, 2009), reaction thermodynamics (Gerstl et al, 2015), participating metabolites (Pey & Planes, 2014), participating reactions (Kaleta et al, 2009), gene regulatory rules (Jungreuthmayer et al, 2013), metabolic subsystems (Schuster et al, 2002; Schwartz et al, 2007), extracellular flux measurements (Soons et al, 2011; Jungers et al, 2011), or even random sampling (Machado et al, 2012). While our HepG2 results did show the majority of source metabolite fluxes were explained by a small number of pathways, contextualizing the significance of these dominant pathway fluxes is only possible when all EFMs are known. Furthermore, reaction thermodynamics, gene regulatory rules, and other constraints may change across fluxes estimated from different biological conditions. Our results suggest that enumerating the shortest pathways would fail to recover the majority of TCA pathways, as the top five ranked EFMs across all glutamine carbons averaged 6 reactions in length. By computing atomic EFMs, instead, we could feasibly enumerate atomic pathways in genome-scale networks containing hundreds of metabolites and reactions within minutes or hours on desktop computing resources. We are hopeful that algorithmic tricks borrowed from molecular EFM enumeration tools, such as linear pathway compression, could further decrease memory usage and run times.

Aside from carbon and nitrogen EFMs, our work is broadly applicable to other types of chemical elements. One could feasibly enumerate oxygen or hydrogen EFMs in small-scale networks, although the significance of these pathways is unclear given the ubiquitous nature of these atoms in metabolic networks. We also caution against enumerating hydrogen EFMs, in particular, given limitations associated with RXNMapper. Unless explicitly represented in the reaction SMILES string, RXNMapper does not assign hydrogen atom mappings. While this may seem trivial, RXNMapper cannot map atoms in reaction SMILES strings longer than 512 characters. Our method could also be applied to study sulfur metabolism to quantify methionine or cysteine remodelling, or phosphorus metabolism to quantify nucleotide synthesis. Outside the realm of biochemistry, our

method may also be relevant to other types of chemical reaction networks and is applicable to any periodic table element.

We foresee the emergence of new bioinformatics tools to analyze atomic EFM weights as technologies to measure high-throughput flux data become increasingly common. While statistical analyses, such as different EFM weight analysis, may reveal metabolic changes in individual pathways altered across conditions, it is likely that changes in a metabolic pathway may impact weights in multiple atomic EFMs. For example, the majority of atomic EFMs across distinct source metabolites are not "molecularly" independent. By this, we imply that some atomic EFMs may correspond to the same sequence of metabolites or correspond to different metabolite/atom sequences stemming from shared reactions. Their weights will likewise be related through these common reactions. This problem may perhaps be addressed by formalizing a relationship between atomic and molecular EFMs. While we have taken significant steps towards solving the EFM enumeration and flux decomposition problem, we expect rapid computational interest analyzing atomic EFM weights in the near future.

Materials and Methods

Reagents and Tools table

See tables/reagents-and-tools-table.xls.

Methods and Protocols

Datasets

Five highly-curated GEMs were chosen in this study to reflect a diversity of organisms, metabolites and biochemical reactions. Four networks were obtained from the BiGG UCSD database (King et al, 2016) in ascending order of network size. Datasets were downloaded as SBML (.xml) files, and the corresponding stoichiometry matrices, metabolites and reactions were extracted using the Julia SBML.jl package. The HepG2 liver cancer cell line dataset was chosen for its single set of estimated steady state fluxes estimated from a combination of protein and metabolite abundance data. These cells were grown in DMEM with 10% fetal calf serum, 22mM glucose, and 1.8mM glutamine (Nilsson et al, 2020b). The steady state fluxes were recomputed from the flux balance analysis (FBA) model provided by the authors (figure3_metabolitebalances.m; MATLAB R2018a) (Nilsson et al 2020b; Data ref: Nilsson et al 2020a). Across all models, we modelled all reversible reactions as net forward reactions to simplify the number of molecular and atomic EFMs as the majority of experimental and computational methods can only estimate net fluxes. We assumed the reaction stoichiometries of the BiGG datasets represented the canonical net directions of all reversible reaction. For the HepG2 dataset, the net directions were chosen based on the net fluxes estimated from the FBA model.

Extracting SMILES strings

SMILES structures, unfortunately, were not encoded in the BiGG SBML files nor the HepG2 dataset. Thus, we manually extracted SMILES strings corresponding to each metabolite across all GEMs. Isomeric SMILES strings were taken from PubChem (Kim et al, 2023), the Human Metabolome Database (Wishart et al, 2022), or MetaNetX database (Moretti et al, 2021). Metabo-

lites shared across GEMs were assigned the same SMILES structure in addition to identical metabolites stratified across distinct subcellular compartments. Generic structures such as R-groups were not modelled in this study and were replaced with a single bonded hydrogen. We assumed this modification would not significantly alter atom mappings, especially since RXNMapper does not map hydrogen atoms unless they are explicitly represented in the SMILES strings. SMILES strings were also not assigned to metabolites with ambiguous structures nor pseudometabolites with no known chemical structure. All pseudometabolites and reactions involving pseudometabolites were removed from the stoichiometry matrix (see Section X for pre-processing details).

Constructing reaction SMILES strings

RXNMapper performs atom mapping on a string that encodes the SMILES structures of all substrates and products for a given reaction. These reaction SMILES strings are constructed by concatenating SMILES strings of individual substrates and products. Multiple substrates or products are separated by the symbol '.', while the reaction arrow delimiting substrates and products is denoted by the symbol '¿¿'. The substrate and product SMILES strings are concatenated in order of metabolite appearance within the stoichiometry matrix, and multiple stoichiometric copies are consecutively repeated in the reaction SMILES strings. Non-integer stoichiometric coefficients are therefore not allowed and these pseudoreactions were removed from the stoichiometry matrix(see Section X for pre-processing details).

Pre-processing genome-scale metabolic models

Several pre-processing steps were required to compute atomic EFMs from the GEMs. The stoichiometric coefficients of external reactions were set to one and their fluxes rescaled to carry equivalent units of source and sink metabolites. All reactions containing pseudometabolites with no known

SMILES strings were removed from the stoichiometry matrix and replaced with unimolecular source and sink reactions carrying equivalent units of flux. The same procedure was also applied to reactions containing non-integer stoichiometries. These pseudometabolites were next removed from the stoichiometry matrix rows, and unimolecular reactions involving the same external or internal metabolites were aggregated. We further split intracellular carbon dioxide metabolites into pairs of sink and source copies to reduce a combinatorial explosion of atomic EFMs. Any reaction producing intracellular carbon dioxide would terminate at a sink carbon dioxide metabolite within that compartment, while any reaction consuming intracellular carbon dioxide would involve a source carbon dioxide metabolite. Finally, we had to remove a small number of reactions whose reaction SMILES strings exceeded the 512 token limit of RXNMapper. These reactions were likewise converted into unimolecular source and sink reactions to maintain steady state flux across the network. We note that all pre-processing steps described above are wrapped into convenience functions within our Julia package MarkovWeightedEFMs.jl.

Benchmarking enumeration of atomic-versus-standard EFM

We benchmarked our Julia package MarkovWeightedEFMs.jl (version X.0.0) running Julia version 1.10.2 to enumerate atomic EFMs versus the molecular EFM enumeration program FluxModeCalculator (van Klinken & Willems van Dijk (2016)) running MATLAB version R2018a with default parameters including network compression. We chose FluxModeCalculator because it is the state-of-the-art program for enumerating molecular EFMs. Benchmarks were performed on a 64-bit Linux operating system with 64GB of memory and an AMD 5950X CPU with 16 cores/32 threads. All benchmarks reported the wall time to run the atomic or molecular EFM enumeration functions and did not include startup times to open Julia/MATLAB, precompile Julia functions, or load packages and input data. A benchmark time of 'did not finish' (DNF) was assigned if either program could

not complete EFM enumeration within 48 hours. In practice, only FluxModeCalculator failed to enumerate molecular EFMs in three of the five datasets.

Model preliminaries

The structure of a knowledge-driven metabolic flux network is encoded as an m by n stoichiometry matrix S with m distinct metabolites and r irreversible reactions. The fluxes along each biochemical reaction v are positive, real-valued weights that denote the number of times each reaction is occurring per unit time. Under metabolic steady state, the net fluxes consuming and producing each metabolite are balanced and $S \cdot v = 0$. These reactions may be unimolecular–single substrate to single product transformations– or involve multiple substrates, products, or stoichiometric copies of participating metabolites. We do not require the network to be strong connected; however, our method will treat independent components separately. In contrast with our previous unimolecular CHMC model, we impose additional requirements on the input metabolites and reactions. The 3D structure of each metabolite must be known and encoded as a SMILES structure. We recommend, where applicable, using isomeric SMILES structures to avoid ambiguous or incorrect atom mappings. We further require integer-valued stoichiometric coefficients because we consider atoms to be indivisible entities from which we enumerate their EFMs. We lastly require the input metabolic flux network to be open such that external fluxes enter and leave via source and sink metabolites. We note this is generally not a problem in large-scale metabolic networks; our requirement is not a technical limitation of our method but enforced to promote the enumeration of source-to-sink EFMs explaining nutrient source remodelling.

A knowledge-driven metabolic network can be decomposed into a finite set of K molecular EFMs. In a unimolecular reaction network, the metabolites and steady state fluxes can be represented as a graph and the molecular EFMs correspond to weighted simple cycles carrying steady

state flux through the metabolite vertices. In a network containing multispecies reactions, the correspondence between molecular EFMs and simple cycles is lost as EFMs may include higher order reactions involving flows across multiple substrates and products. For either type of knowledge-driven metabolic network, their observed steady state fluxes can always be represented as a positive, linear combination of their K molecular EFMs. This flux decomposition problem can be represented by the system of linear equations $Aw = f$, where A is an r by K matrix of zero or natural numbers denoting the number of times each reaction occurs molecular EFM $k = 1, 2, \ldots, K$, where w_k is the molecular EFM weight. When the number of molecular EFMs exceeds the number of linearly independent fluxes, w is underdetermined and there are an infinite number of ways to assign molecular EFM weights to explain the observed, steady state fluxes.

Markovian decomposition of fluxes onto EFM weights

In our previous work (Chitpin & Perkins, 2023), we proposed Markovian EFM weights as a means to resolve the flux decomposition problem in networks consisting exclusively of unimolecular reactions. We formulated a probabilistic model of single particle flux, in the form of a Markov chain, where we assumed the probability of this hypothetical particle transitioning from one metabolite state to another was proportional to the observed, steady state flux. We further proposed a way of efficiently computing those molecular EFM probabilities and weights by introducing the *cycle-history Markov chain* which both enumerated the molecular EFMs and uniquely identified their probabilities which were rescaled to weights based on the absolute flux magnitudes. To generalize this CHMC method to networks containing multispecies reactions, we introduce the concept of atomic EFMs which can be enumerated and their weights computed from an atomic variant of our CHMC method. The following sections provide intuitive justification of our method, while we leave our proof of (atomic/single particle) CHMC algorithm correctness in our original paper (Chitpin & Perkins,

2023).

Single atom, discrete-time Markov chain

We developed a probabilistic model of atomic flux to compute atomic EFM weights in steady state metabolic flux networks. At this atomic level, individual atoms within metabolite undergo biochemical reactions and move from the substrate to product side of each reaction equation. As the molecules within metabolic flux network are stoichiometrically balanced under metabolic steady state, so too are the corresponding atomic fluxes.

For a user-specified atom within a given source metabolite, we identify all possible metabolite/atom states that are traversable by that initial state. For reactions involving single stoichiometric copies of each substrate and product, the atom of interest will always move from a single substrate to product molecule. If this atom is present in a substrate with multiple stoichiometric copies, we assume there is an equal probability of that atom occurring in either copy. Depending on the substrate stoichiometric copy, that atom may move to (i) different product metabolite/atom states, (ii) the same product but at different atomic positions, or (iii) the same metabolite/atom position. In all cases, we set the atomic flux of the atom transitioning from either stoichiometric copy equal to the reaction fluxes to maintain atomic mass conservation.

For a given source metabolite/atom state, the subsequent atomic transitions are encoded as an atomic transition graph $G = (V, E, f)$, where V is the set of nodes corresponding to metabolite/atom states, E is the set of directed edges corresponding to reactions that move the atom from one metabolite to another, and $f \equiv v$ are the atomic mass fluxes. We assume there exists a special node $v_{ext} \in V$ which represents the external environment. Source flux entering the graph occurs along the edge $e = (v_{ext}, r)$, where $r \in V - v_{ext}$ is the source metabolite/atom state. Sink fluxes leaving the graph occur along edges $e = (i, v_{ext})$, where $i \in V - v_{ext}$ are the sink metabolite/atom states.

The introduction of v_{ext} closes off the graph and ensures it is strongly connected, where there exists a path in G from any node i to any other node j.

We make a Markovian assumption that the movement of an atom from metabolite/atom position to another in G is proportional to the observed steady state (atomic) flux of that reaction. We denote these transition probabilities $T_{ij} = f_{ij}/\sum_{j'} f_{ij'}$ for $i \neq j \in |V|$. From here, we follow the same intuition as our unimolecular CHMC algorithm (Chitpin & Perkins, 2023) to model atomic flows within the network. Starting from an initial source metabolite/atom state s_0, the atom of interest will transit a random sequences of states $s_0, s_1, s_2, \ldots$. As G is strongly connected, that sequence will, with probability of one, include all states infinitely many times. If the atom visits state i, it will eventually return to state i. The sequence of states in between $s_i, \ldots, s_{i'}$ constitutes a cycle, albeit not necessarily a simple cycle. The sequence many include multiple nested or interleaved simple cycles. However, this cycle can always be decomposed into a unique set of simple cycles by repeatedly removing the earliest simple cycle found within the sequence, which we refer to as the "first closure reduction rule". The atomic EFMs are the set of all simple cycles in G. Thus, we compute the atomic EFM weights by setting them proportional to the steady state probabilities of their corresponding simple cycles.

Atomic cycle-history Markov chain

We compute the steady state probabilities of each atomic EFM following our previously described CHMC method for networks of unimolecular reactions (Chitpin & Perkins, 2023). Briefly, the atomic CHMC is a special type of Markov chain, where we expand the notion of state to include sequences of metabolites transited by the atom. The atomic CHMC therefore traces the history of a given atom on its way to completing a simple cycle corresponding to an atomic EFM.

We first initialize the atomic CHMC on source metabolite atom s_0 within G. Although we could

technically choose any arbitrary metabolite/atom state for s_0, it is much easier to interpret atomic EFMs rooted on a source metabolite flowing into the network. Every state S in the atomic CHMC corresponds to a simple path $s_0, s_1, s_2, \ldots$. If a reaction transforms s_m to s_{m+1}, the corresponding state $S \equiv s_0, s_1, \ldots, s_m$ transitions to the longer path $S' \equiv s_0, s_1, \ldots, s_m, s_{m+1}$. If a reaction transforms $s_{m+1} = s_i$, the corresponding state $S \equiv s_0, s_1, \ldots, s_m$ transitions to the shorter path $S' \equiv s_0, s_1, \ldots, s_i$ For either case, the atomic CHMC transition probability is $\mathcal{T}_{S,S'} = T_{s_m, s_{m+1}}$.

Computing steady state atomic EFM probabilities

Once constructed, the atomic CHMC describes all possible simple cycles that a given source metabolite atom can transit through. These simple cycle probabilities are computed by analyzing the steady state dynamics of the atomic CHMC. As the atomic transition graph G is strongly connected, the corresponding CHMC will also be strongly connected. There thus exists a stationary distribution π which corresponds to the steady state probability of the atom occupying each CHMC state.

To compute atomic EFM probabilities, we first identify all transitions E_K in the atomic CHMC that close simple cycles corresponding to EFM k. For source-to-sink atomic EFMs, there exists only a single simple cycle and therefore only one corresponding simple cycle. For internally looped atomic EFMs, there may exist multiple simple cycles starting at different metabolite/atom states, counting towards the same atomic EFM. We define the steady state probability of transiting EFM k as the sum of the steady state probabilities of those transitions or:

$P_k = \sum_{(S,S') \in E_k} \pi_S \cdot \mathcal{T}_{S,S'}$.

As the atomic EFM probabilities are proportional to the atomic fluxes, these probabilities may be scaled to weights based on the absolute magnitude of the fluxes. Because we enforce that each source metabolite is produced by an input reaction with stoichiometry of one, the sum of all corresponding

source-to-sink atomic EFMs must be proportional to the input metabolite flux. If $l \in K$ denotes all source-to-sink atomic EFMs flowing through a given source metabolite m_0 and f_0 is the input flux producing that metabolite, we define the proportionality constant $\alpha = f_0 / \sum_l P_l$. This scaling not only ensures the total input/output fluxes are correctly reconstructed, but also that every individual flux in the network is correctly explained by the totality of atomic EFM weights.

Details associated with computing the CHMC stationary distribution

Following basic Markov chain theory, the stationary distribution of the CHMC, π is proportional to the eigenvector of the CHMC transition probability matrix. The eigenvector can be solved by direct or iterative numerical methods. The choice of algorithm becomes increasingly important as the size of the CHMC transition matrix grows exponentially in highly-connected, large-scale networks. We note that iterative eigenvector solving algorithms from some packages (e.g. Arnoldi method from Arnoldi.jl, GMRES from IterativeSolvers.jl) resulted in negative eigenvector coefficients or NaN values. These are possibly numerical errors and occurred more frequently in larger CHMC matrices. Repeatedly re-running these methods sometimes resulted in a correct eigenvector, albeit with significantly longer computation times. We observed that the linear solver from the Julia package LinearSolve.jl efficiently solved for the correct eigenvector without error using the default parameters. Hence, we use and recommend this package to compute the stationary distribution of large atomic CHMCs.

Data availability

The code and data to reproduce this study are available at

https://www.github.com/jchitpin/reproduce-efm-paper-2024.

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Author contributions

Justin G. Chitpin: Conceptualization; data curation; methodology; software; formal analysis; investigation; visualization; writing - original draft; writing - review and editing.

Theodore J. Perkins: Conceptualization; methodology; supervision; funding acquisition; investigation; project administration; writing - review and editing.

Disclosure and competing interests statement

The authors declare no competing interests.

References

Ahn E, Kumar P, Mukha D, Tzur A, Shlomi T (2017) Temporal fluxomics reveals oscillations in TCA cycle flux throughout the mammalian cell cycle. *Mol Syst Biol* 13: 953

Antoniewicz MR (2018) A guide to 13C metabolic flux analysis for the cancer biologist. *Exp Mol Med* 50: 1

Arnold PK, et al (2022) A non-canonical tricarboxylic acid cycle underlies cellular identity. *Nature* 603: 477–481

Buchner BA, Zanghellini J (2021) EFMlrs: a python package for elementary flux mode enumeration via lexicographic reverse search. *BMC Bioinformatics* 22: 1–547

Buescher JM, et al (2015) A roadmap for interpreting 13C metabolite labeling patterns from cells. *Curr Opin Biotechnol* 34: 189–201

Cherkaoui S, et al (2022) A functional analysis of 180 cancer cell lines reveals conserved intrinsic metabolic programs. *Mol Syst Biol* 18: e11033

Chitpin JG, Perkins TJ (2023) A Markov constraint to uniquely identify elementary flux mode weights in unimolecular metabolic networks. *J Theor Biol* 575: 111632–111632

de Figueiredo LF, Podhorski A, Rubio A, Kaleta C, Beasley JE, Schuster S, Planes FJ (2009) Computing the shortest elementary flux modes in genome-scale metabolic networks. *Bioinformatics* 25: 3158–3165

Emwas AH, Szczepski K, Al-Younis I, Lachowicz JI, Jaremko M (2022) Fluxomics - new metabolomics approaches to monitor metabolic pathways. *Front Pharmacol* 13: 805782–805782

Gagneur J, Klamt S (2004) Computation of elementary modes: a unifying framework and the new binary approach. *BMC Bioinformatics* 5: 175–175

Gelbach PE, Cetin H, Finley SD (2024) Flux sampling in genome-scale metabolic modeling of microbial communities. *BMC Bioinformatics* 25: 45

Gerstl MP, Ruckerbauer DE, Mattanovich D, Jungreuthmayer C, Zanghellini J (2015) Metabolomics integrated elementary flux mode analysis in large metabolic networks. *Sci Rep* 5: 8930

González-Arrué N, Inostroza I, Conejeros R, Rivas-Astroza M (2023) Phenotype-specific estimation of metabolic fluxes using gene expression data. *iScience* 26: 106201–106201

Huang Y, Mohanty V, Dede M, Tsai K, Daher M, Li L, Rezvani K, Chen K (2023) Characterizing cancer metabolism from bulk and single-cell rna-seq data using METAFlux. *Nat Commun* 14: 4883–4883

Jungers RM, Zamorano F, Blondel VD, Wouwer AV, Bastin G (2011) Fast computation of minimal elementary decompositions of metabolic flux vectors. *Automatica (Oxf)* 47: 1255–1259

Jungreuthmayer C, Ruckerbauer DE, Zanghellini J (2013) regEfmtool: Speeding up elementary flux mode calculation using transcriptional regulatory rules in the form of three-state logic. *BioSystems* 113: 37–39

Kaleta C, de Figueiredo LF, Schuster S (2009) Can the whole be less than the sum of its parts? pathway analysis in genome-scale metabolic networks using elementary flux patterns. *Genome Res* 19: 1872–1883

Kamp Av, Schuster S (2006) Metatool 5.0: fast and flexible elementary modes analysis. *Bioinformatics* 22: 1930–1931

Kaste JAM, Shachar-Hill Y (2023) Accurate flux predictions using tissue-specific gene expression in plant metabolic modeling. *Bioinformatics* 39

Kim S, et al (2023) PubChem 2023 update. *Nucleic Acids Res* 51: D1373–D1380

King ZA, Lu J, Dräger A, Miller P, Federowicz S, Lerman JA, Ebrahim A, Palsson BO, Lewis NE (2016) BiGG Models: a platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Res* 44: D515–D522

Lee JY, Han Y, Styczynski MP (2023) Towards inferring absolute concentrations from relative abundance in time-course gc-ms metabolomics data. *Mol Omics* 19: 126–136

Machado D, Soons Z, Patil KR, Ferreira EC, Rocha I (2012) Random sampling of elementary flux modes in large-scale metabolic networks. *Bioinformatics* 28: i515–i521

Moretti S, Tran V, Mehl F, Ibberson M, Pagni M (2021) MetaNetX/MNXref: unified namespace for metabolites and biochemical reactions in the context of metabolic models. *Nucleic Acids Res* 49: D570–D574

Nilsson A, Haanstra JR, Engqvist M, Gerding A, Bakker BM, Klingmüller U, Teusink B, Nielsen J (2020a) GitHub (https://github.com/sysbiochalmers/livercellmetabolismsimulation). [DATASET]

Nilsson A, Haanstra JR, Engqvist M, Gerding A, Bakker BM, Klingmüller U, Teusink B, Nielsen J (2020b) Quantitative analysis of amino acid metabolism in liver cancer links glutamate excretion to nucleotide synthesis. *Proc Natl Acad Sci U S A* 117: 10294–10304

Orth JD, Fleming RMT, Palsson BØ (2010) Reconstruction and use of microbial metabolic networks: the core Escherichia coli metabolic model as an educational guide. *Ecosal Plus* 4

Pey J, Planes FJ (2014) Direct calculation of elementary flux modes satisfying several biological constraints in genome-scale metabolic networks. *Bioinformatics* 30: 2197–2203

Schuster S, Dandekar T, Fell DA (1999) Detection of elementary flux modes in biochemical networks: a promising tool for pathway analysis and metabolic engineering. *Trends Biotechnol* 17: 53–60

Schuster S, Hilgetag C (1994) On elementary flux modes in biochemical reaction systems at steady state. *J Biol Syst* 2: 165–182

Schuster S, Pfeiffer T, Moldenhauer F, Koch I, Dandekar T (2002) Exploring the pathway structure of metabolism: decomposition into subnetworks and application to mycoplasma pneumoniae. *Bioinformatics* 18: 351–361

Schwaller P, Hoover B, Reymond JL, Strobelt H, Laino T (2021) Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Adv* 7: eabe4166

Schwartz JM, Gaugain C, Nacher JC, de Daruvar A, Kanehisa M (2007) Observing metabolic functions at the genome scale. *Genome Biology* 8: R123–R123

Schwartz JM, Kanehisa M (2006) Quantitative elementary mode analysis of metabolic pathways: the example of yeast glycolysis. *BMC Bioinformatics* 7: 186–186

Soons ZI, Ferreira EC, Rocha I (2011) Identification of minimal metabolic pathway models consistent with phenotypic data. *J Process Control* 21: 1483–1492

Terzer M, Stelling J (2008) Large-scale computation of elementary flux modes with bit pattern trees. *Bioinformatics* 24: 2229–2235

Uematsu S, et al (2022) Multi-omics-based label-free metabolic flux inference reveals obesity-associated dysregulatory mechanisms in liver glucose metabolism. *iScience* 25: 103787–103787

Ullah E, Yosafshahi M, Hassoun S (2020) Towards scaling elementary flux mode computation. *Brief Bioinform* 21: 1875–1885

van Klinken JB, Willems van Dijk K (2016) FluxModeCalculator: an efficient tool for large-scale flux mode computation. *Bioinformatics* 32: 1265–1266

Wishart DS, et al (2022) HMDB 5.0: the Human Metabolome Database for 2022. *Nucleic Acids Res* 50: D622–D631

Wu C, Herold RA, Knoshaug EP, Wang B, Xiong W, Laurens LML (2019) Fluxomic analysis reveals central carbon metabolism adaptation for diazotroph Azotobacter vinelandii ammonium excretion. *Sci Rep* 9: 13209–11

Zamboni N, Saghatelian A, Patti G (2015) Defining the metabolome: Size, flux, and regulation. *Mol Cell* 58: 699–706

Figure legends

Figure 1: Atomic cycle-history Markov chain pipeline. (A) An example molecular EFM in glycolysis. (B) An example atomic EFM tracing the flow of a given glucose carbon in glycolysis. (C) The major steps to enumerate atomic EFMs and compute their weights.

Figure 2: Atomic EFMs are computationally tractable compared to molecular EFMs in five genome-scale metabolic models. (A) Number of atomic and molecular EFMs in the five genome-scale models. (B) The run time of our MarkovWeightedEFMs.jl method versus FluxModeCalculator running serially and in parallel. (C) Run time scaling of MarkovWeightedEFMs.jl as a function of the number of atomic EFMs. (D) Speedup of selected atomic CHMCs as a functional of additional threads.

Figure 3: Structural analysis of atomic EFMs in five genome-scale metabolic models. (A-B) Categories of carbon and nitrogen EFMs. (C-D) Distribution carbon and nitrogen EFM lengths. (E-F) Ratio of the greatest and smallest number of EFMs between carbons and nitrogens within the same source metabolite.

Figure 4: Carbon EFM weight analysis of the HepG2 metabolic flux network. (A) A small number of highly active EFMs explain the majority of glutamine flow. (B) Subnetwork of the top five EFMs across each glutamine carbon collapsed onto metabolites. (C-F) Sankey diagrams of the top five

glutamine carbons.

Tables and their legends

Table 1: Summary statistics of five genome-scale metabolic models.

Expanded View Figure legends

Expanded Views 1: FluxModeCalculator scaling across multiple threads.

Expanded Views 2: Cumulative explained carbon mass flow of input glucose and amino acids from the HepG2 model.